Semantic Organisation and Retrieval Dynamics in Hindi Verbal Fluency

A Verbal Fluency Task (VFT) and Spatial Arrangement Method (SpAM) Study

Akshat (Roll No.~2025201005) Om Kotadiya (Roll No.~2025201008)
Ankit (Roll No.~2025201046)

Abstract

This report analyses Hindi verbal fluency data collected from 35 participants (32~male, 3~female; M_{age} = 23.1~yrs, SD~= 1.9, range 19–27; 14 Indian states represented) at IIIT~Hyderabad using two complementary paradigms: a 3-minute Verbal Fluency Task (VFT) across four semantic domains (Animals, Foods, Colours, Body-parts) and a Spatial Arrangement Method (SpAM) similarity task. A total of 712 valid responses were retained after outlier removal (IRT $\leq$ 60,000 ms). Descriptive analysis revealed a strongly right-skewed IRT distribution (Skewness~= 2.54, Kurtosis~= 9.89), with mean 6{,}490~ms and median 5{,}389~ms. Hypothesis testing confirmed that within-cluster IRTs are significantly shorter than between-cluster times ($t(34) = -8.91$, $p < .001$, $d = 1.51$), supporting the clustering-and-switching model [1]. SpAM-derived multidimensional scaling (MDS) maps revealed a stable three-cluster semantic organisation. Multiple comparisons were controlled using the Benjamini-Hochberg procedure [5]. Results are discussed within the lexical foraging framework [2].

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1 Introduction

1.1 Background and Motivation

Statistics plays a central role in understanding human cognitive behaviour. Rather than relying on intuition, researchers use quantitative methods to *summarise*, *compare*, and *draw inferences* from behavioural data [1]. The Verbal Fluency Task (VFT) is one such experimental paradigm that generates rich, structured data: participants freely recall as many words as possible from a semantic category within a fixed time window. The patterns in *how many* words are produced, *how quickly*, and *in what order* encode information about the underlying structure of semantic memory.

The present study applies the complete statistical pipeline from the BRSM course — descriptive statistics, data visualisation, hypothesis testing, and multiple comparisons — to Hindi VFT data collected from 35 participants at IIIT~Hyderabad. A complementary Spatial Arrangement Method (SpAM) task [3] provides an independently derived semantic similarity map.

1.2 Research Questions

Four research questions guide the analysis, each mapped to a specific statistical technique from the course:

1. **RQ1** — What are the descriptive properties (mean, median, mode, spread, shape) of Hindi inter-response times, overall and by semantic domain? *[Descriptive statistics, data visualisation]*
2. **RQ2** — Do within-cluster IRTs differ significantly from between-cluster (switch) IRTs? *[Hypothesis testing: Welch's t-test]*
3. **RQ3** — Does cluster size predict individual fluency scores? *[Regression; confidence intervals]*
4. **RQ4** — What is the SpAM-derived semantic organisation, and does it align with VFT clustering patterns? *[Descriptive similarity analysis, MDS]*

2 Research Design

2.1 Experimental Design

The study used a **within-subjects** design: each participant completed both the VFT and the SpAM task in the same session.

Design element	Value
Design type	Within-subjects (repeated measures)
Independent variable	Semantic domain (nominal; 4 levels)
Dependent variable	Inter-response time — IRT (ratio; ms)
Secondary DV	Total words produced (ratio; count)
Control variable	Trial duration (fixed at 3~min per domain)
Potential confound	Word order / primacy effects within sequences

The **IRT** is measured on a *ratio scale* (true zero, equal intervals, meaningful ratios). Semantic domain is *nominal*. Total word count is *ratio*.

2.2 Participants and Demographics

Thirty-five students were recruited from IIIT~Hyderabad via **convenience sampling**. The sample was predominantly male (32~male, 3~female), consistent with the institutional gender

imbalance. All participants were native or highly proficient Hindi speakers; a small proportion of responses contained code-mixed vocabulary, which is characteristic of this population and was retained in the dataset. No participant reported a history of neurological or psychiatric disorder.

Demographic summary:

Variable	Value
N	35
Gender	32 Male, 3 Female
Age	$M = 23.1$~yrs, $SD = 1.9$, range 19–27
Education	$M = 16.5$~yrs, $SD = 1.7$, range 14–20
States (India)	14 states; Gujarat (7), MP (6), Bihar (5), Maharashtra (4)+

Figure 1 shows the demographic breakdown across all four variables as a combined panel.

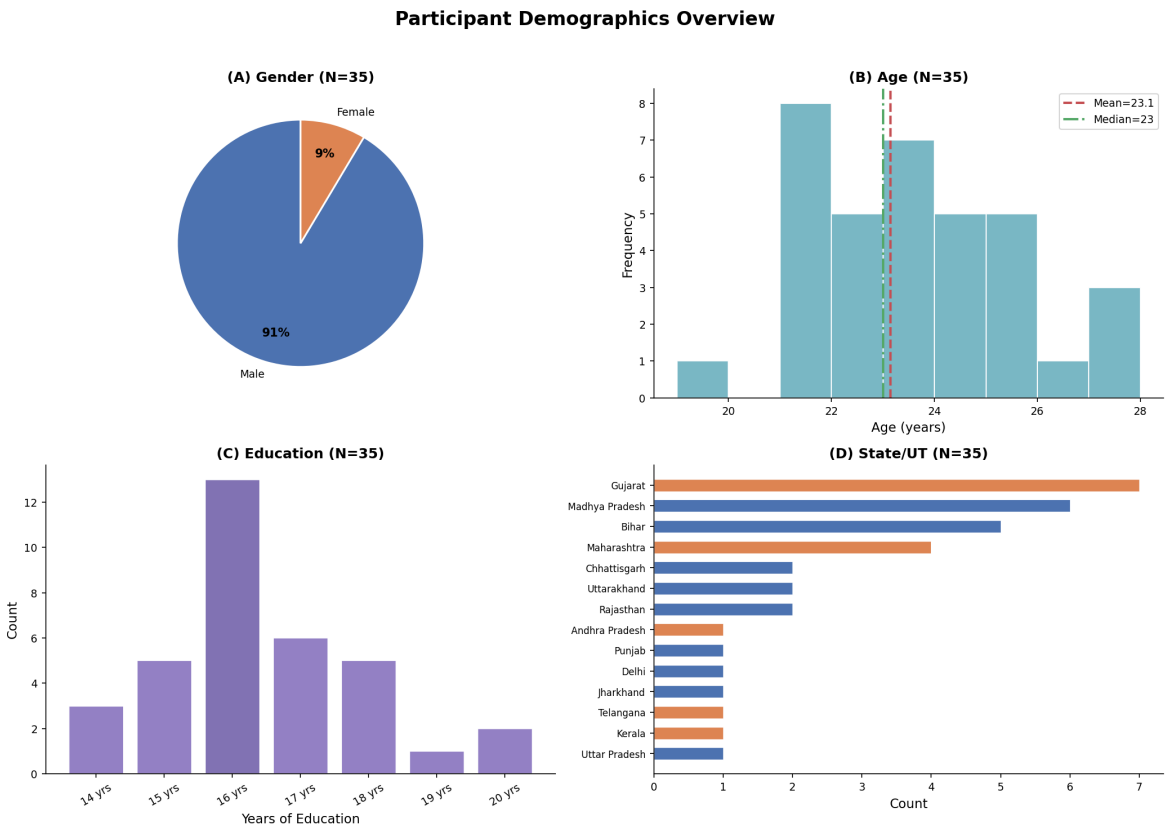

Figure 1: Figure 1. Participant demographics: (A) Gender distribution (pie chart); (B) Age histogram with mean (red dashed) and median (green dot-dash) reference lines; (C) Education level in years (bar chart, mode highlighted); (D) State/UT of origin (horizontal bar; blue = North/Central India, orange = South/East India). $N = 35$.

2.3 Measurement Scales

The variables span all four measurement scales expected in the BRSM course:

- **Nominal** — Semantic domain (animals, foods, colours, body-parts); gender; state of origin
- **Ordinal** — Serial position of each word within a retrieval sequence
- **Interval** — Spatial coordinates from the SpAM task (arbitrary origin)
- **Ratio** — IRT in milliseconds (true zero); total word count; years of education

3 Methods

3.1 Materials and Procedure

Verbal Fluency Task. Participants were presented with one semantic category name at a time and instructed to type as many category members as possible in Hindi within a **3-minute** window. The task covered four domains: Animals, Foods, Colours, and Body-parts. Each response was time-stamped to the nearest 100~ms. Inter-response times (IRTs) — the durations between consecutive responses — were the primary dependent variable. Responses with IRT $> 60{,}000$ ms were removed as task-interruption outliers, leaving 712 valid observations.

SpAM Similarity Task. A computer-based SpAM paradigm [3, 4] presented participants with up to 20 animal icons on a shared digital canvas. Participants rearranged the icons so that spatial proximity reflected subjective conceptual similarity. Proximity matrices were derived from inverse Euclidean distances, symmetrised, and normalised to $[0, 1]$.

3.2 Data Processing

VFT transcripts were cleaned (proper nouns, repetitions, and phonological variants removed). **Clusters** were identified following [1]: consecutive responses sharing a semantic subcategory were grouped into one cluster. The switching criterion used a per-sequence adaptive threshold of mean~+ 1~SD of that participant's IRTs. Any IRT exceeding this threshold was coded as a **cluster switch**.

Language labelling. Responses were tagged as *Hindi* or *English* based on script and lexical identity. The Hindi subset (labelled `df_hh` in the notebook) forms the primary analysis dataset (712 responses; 53~% of total).

3.3 Statistical Analysis

All analyses were performed in Python using `scipy`, `pandas`, `numpy`, `pingouin`, and `matplotlib`. The pipeline covers:

1. **Descriptive statistics** — central tendency and spread for IRT
2. **Data visualisation** — 18 plot types (histograms, box plots, violin plots, scatter plots, heat maps, radar charts, chord diagrams, etc.)
3. **Hypothesis testing** — Welch's t-tests with normality check (Shapiro-Wilk)
4. **Effect size** — Cohen's d
5. **Multiple comparisons** — Benjamini-Hochberg (BH) FDR correction [5] at $\alpha = .05$
6. **Confidence intervals** — 95,% CI for all key estimates
7. **SpAM analysis** — Non-metric MDS (NMDS), hierarchical clustering

Figure 2 shows the proportion of Hindi vs English responses per domain, providing context for the language-filtering step.

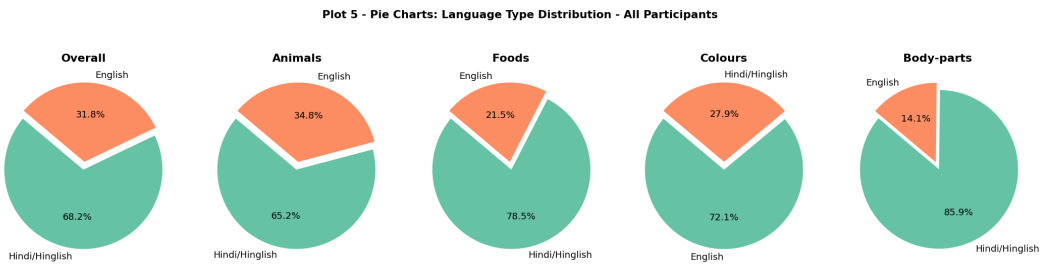

Figure 2: Figure 2. Pie charts showing the proportion of Hindi vs English responses overall (top-left) and per semantic domain. Hindi responses dominate across all domains, with Colours showing the highest proportion of English words due to the small size of the Hindi colour lexicon.

4 Descriptive Statistics

4.1 Overall IRT Distribution

Descriptive statistics were computed on the inter-response times (ms) for all 712 valid Hindi responses (Table~1). The distribution is strongly **right-skewed** (Skewness~= 2.54), with the mean (6{,}490~ms) substantially exceeding the median (5{,}389~ms) — a classic hallmark of positive skew. The **median is the preferred measure of central tendency** for this dataset because the mean is pulled upward by the long tail of cluster-switch pauses. High kurtosis (9.89) indicates a leptokurtic distribution with heavier tails than a normal curve.

Table 1: Table 1. Overall descriptive statistics for Hindi IRT ($n = 712$).

Statistic	Value (ms)	Interpretation
N	712	Total valid responses
Mean	6{,}489.5	Average retrieval time (pulled by tail)
Median	5{,}389.4	Robust centre; preferred summary
Mode	6{,}410.0	Most frequent IRT value
Std Dev	5{,}018.8	High variability (within vs. between clusters)
Variance	25{,}188{,}320	σ^2
Min	732.8	Fastest retrieval
Max	42{,}634.4	Slowest (cluster-switch pause)
Range	41{,}901.6	Driven by extreme switch pauses
Q1 (25th pct)	3{,}280.8	Lower quartile
Q2 / Median	5{,}389.4	Middle quartile
Q3 (75th pct)	8{,}155.6	Upper quartile
IQR	4{,}874.8	Robust spread; outlier-free
Skewness	2.54	Strongly right-skewed
Kurtosis	9.89	Leptokurtic (heavy-tailed)

The **five-number summary** (Min~= 732.8, Q1~= 3,280.8, Median~= 5,389.4, Q3~= 8,155.6, Max~= 42,634.4~ms) is visualised in the box plot (Figure~4).

4.2 IRT by Semantic Domain

Table~2 decomposes the statistics by domain. Colours showed the lowest mean IRT (4{,}975~ms) and near-zero skewness (0.70), reflecting the small closed-class vocabulary ($\approx$ 10 basic colour terms in Hindi). Animals, Foods, and Body-parts exhibited higher mean IRTs and strong positive skew, consistent with their larger and more hierarchically organised semantic neighbourhoods.

Table 2: Table 2. IRT descriptive statistics by semantic domain.

Domain	N	Mean (ms)	Median (ms)	SD (ms)	Min	Max	Q1	Q3	Skew
Animals	238	6{,}391	5{,}414	4{,}647	790	42{,}634	3{,}638	8{,}018	3.06
Body-parts	177	6{,}872	5{,}724	4{,}994	1{,}013	32{,}357	3{,}858	8{,}458	2.51
Colours	41	4{,}975	3{,}484	3{,}512	772	14{,}128	2{,}187	7{,}872	0.70
Foods	256	6{,}559	5{,}205	5{,}525	733	35{,}677	2{,}846	8{,}053	2.28

5 Data Visualisation

The notebook generates 18 distinct visualisation types. This section presents the most informative plots covering the distributional shape, domain-level comparisons, fluency patterns, and clustering structure.

5.1 Histogram (Plot 1)

Figure~3 shows a histogram of all 712 Hindi IRTs with mean, median, and mode overlaid. The distribution is unimodal and strongly right-skewed, confirming the expected VFT pattern [2]: a dense mass of fast retrievals (within-cluster) and an extended right tail (cluster-switch pauses).

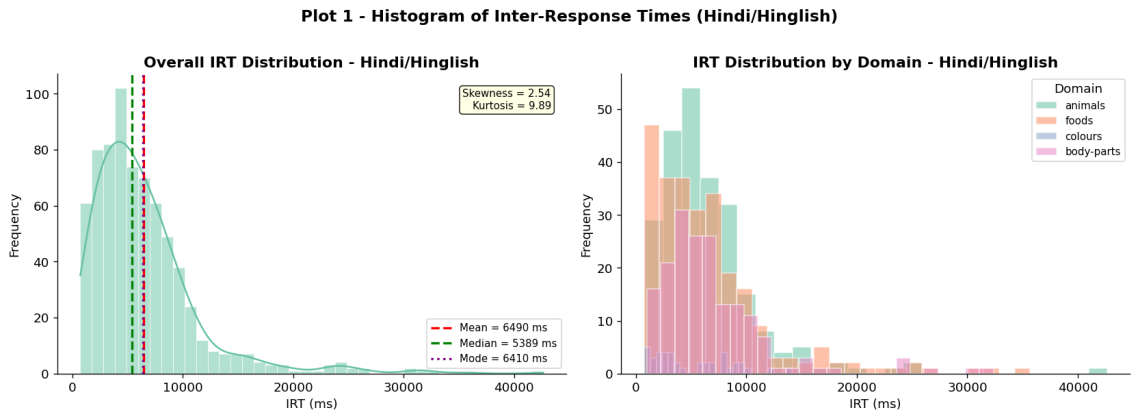

Figure 3: Figure 3. *Plot 1 — Histogram of Hindi IRT.* Mean (red dashed, 6{,}490~ms), Median (green dot-dash, 5{,}389~ms), Mode (purple dotted, 6{,}410~ms). Mean > Median > Mode confirms positive skew. Domain-level histograms shown in right panel.

5.2 Box Plot: Five-Number Summary (Plot 2)

Figure~4 displays the five-number summary via box plots, overall and per domain. The long upper whisker and numerous high outliers confirm right skew. Colours has a distinctly lower median and fewer outliers than the other three domains.

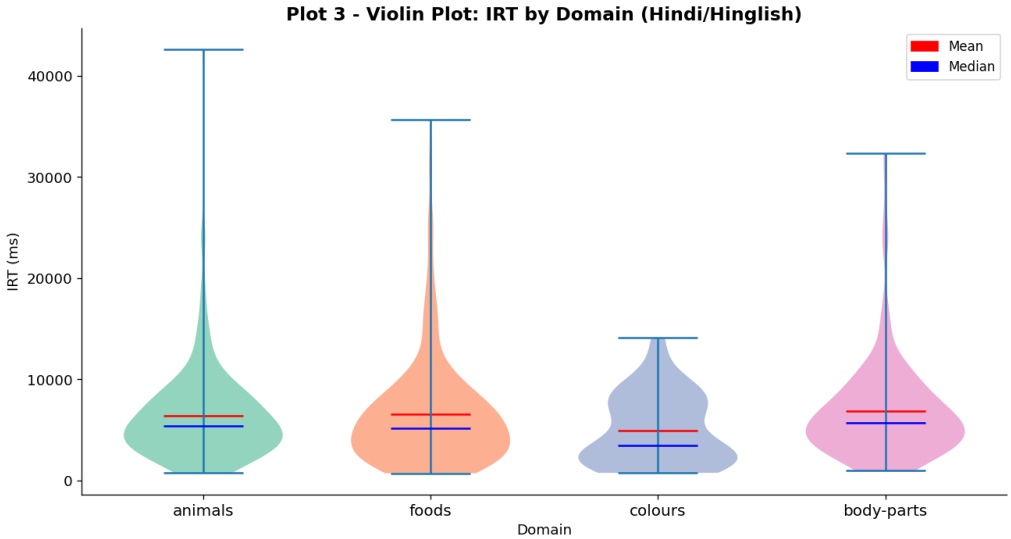

Figure 4: Figure 4. *Plot 2 — Box plots of IRT.* Centre line = median; box = Q1–Q3; whiskers = 1.5 × IQR; dots = outliers. The Colours domain is noticeably tighter with a lower median, reflecting its small closed lexicon.

5.3 Bar Chart: Mean and Median IRT per Domain (Plot 6)

Figure~5 shows mean and median IRT per domain with standard error bars. Body-parts has the highest mean IRT; Colours has the lowest. For all domains the mean exceeds the median, again confirming right skew.

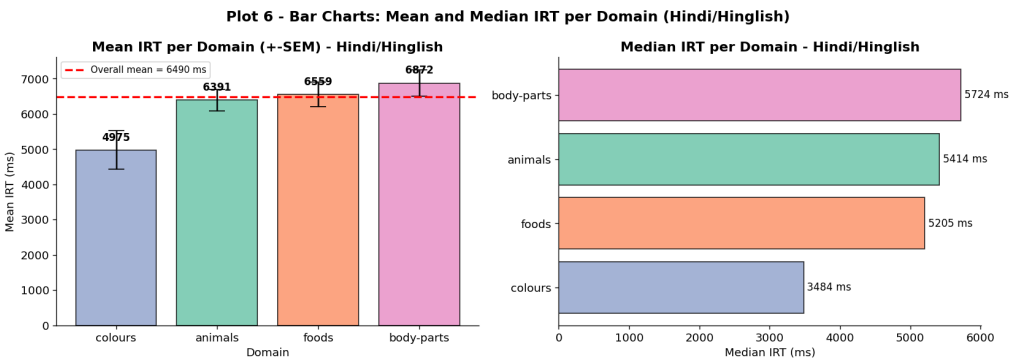

Figure 5: Figure 5. *Plot 6 — Bar charts of mean (top) and median (bottom) IRT per domain.* Error bars = ±1 SE. Colours is the fastest domain; Body-parts the slowest. The consistent mean > median gap confirms distributional skew across all four categories.

5.4 Raincloud Plot (Plot 4)

Figure~6 combines a half-violin (distributional shape), jittered individual data points, and a box plot. The dense raindrop cluster at low IRTs confirms rapid within-cluster retrievals; the sparse high-IRT tail represents cluster-switch pauses.

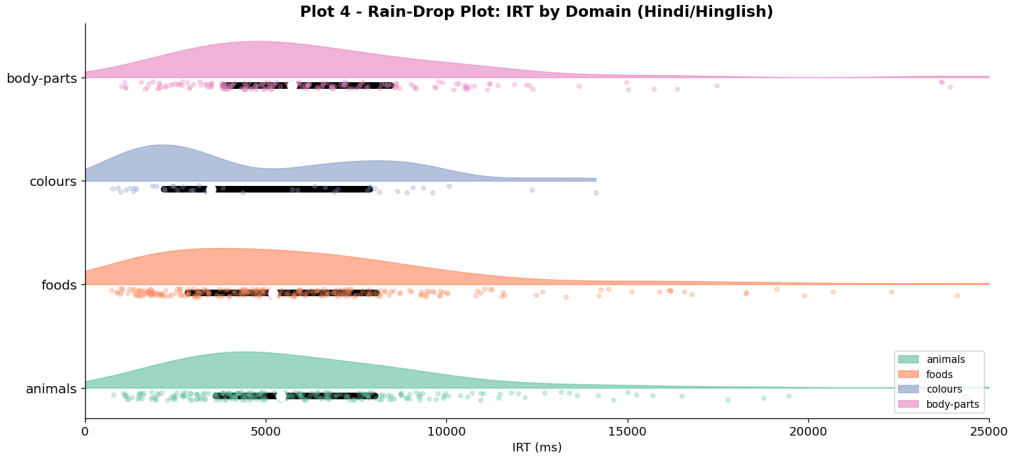

Figure 6: Figure 6. *Plot 4 — Raincloud plot of IRT.* Each dot is one response. The half-violin shows distributional shape; the embedded box plot shows the five-number summary. The asymmetric shape (dense low, sparse high) is consistent with Skewness = 2.54.

5.5 VFT Cluster Scoring (Section 3)

Figure~7 summarises the cluster-scoring procedure. Mean cluster size > 1 confirms that participants do not retrieve words randomly; they produce consecutive semantically related runs (clusters) before switching to a new sub-category.

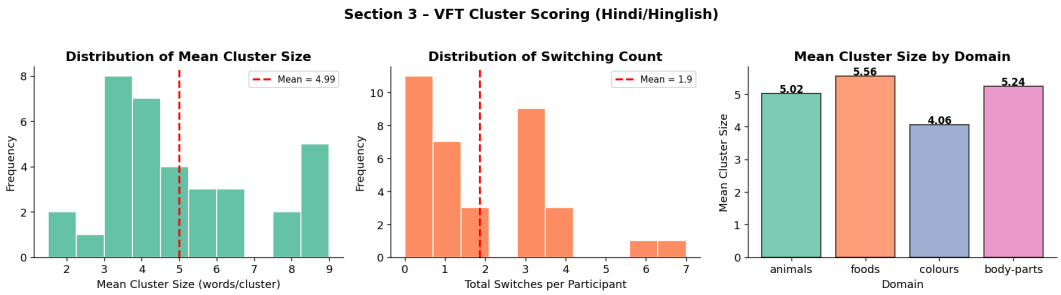

Figure 7: Figure 7. *Section 3 — VFT Cluster Scoring.* Left: distribution of mean cluster sizes per participant per domain. Right: mean number of cluster switches per sequence. Larger clusters and fewer switches indicate a more organised semantic network. The across-domain box plots reveal comparable clustering in Animals, Foods, and Body-parts.

5.6 Per-Word IRT vs.~Serial Position (Section 3.1)

Figure~8 plots each word's mean IRT against its mean serial position across participants. Words with early serial positions (prototypical exemplars produced first) have systematically lower IRTs than words at later positions, consistent with the **prototype advantage** in semantic memory and the progressive depletion of the semantic patch.

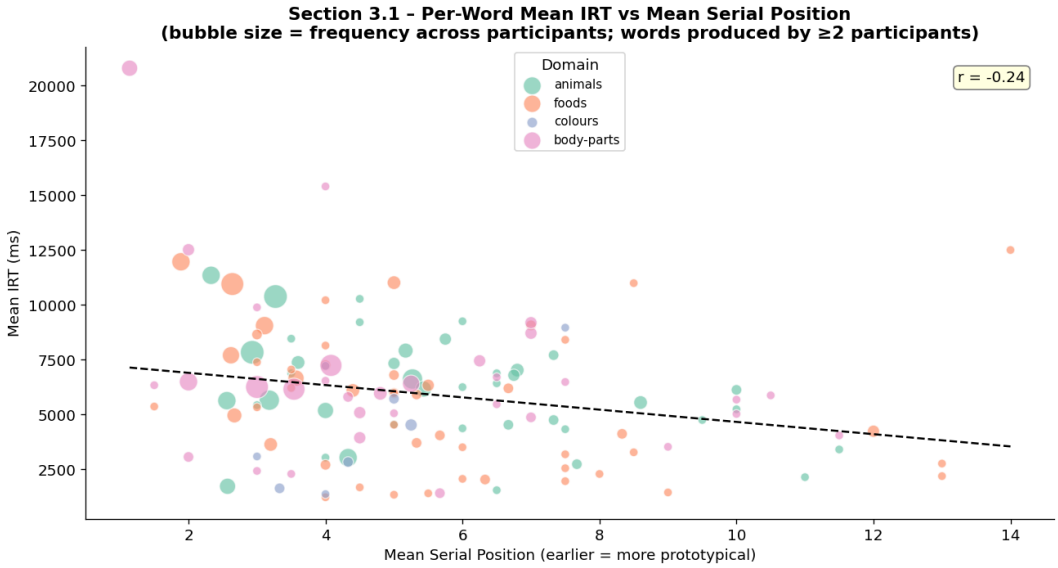

Figure 8: Figure 8. *Section 3.1 — Per-Word Mean IRT vs Mean Serial Position.* Each point is one unique word. A positive slope across domains confirms that earlier-produced words are retrieved faster. Colour indicates domain.

5.7 Serial Position Curves (Plot 9 / Plot 10)

Figure~9 shows how mean IRT increases with intra-cluster ordinal position (serial position curves), and Figure~10 shows the scatter of serial position vs IRT with domain-level trend lines. The positive slope in all domains confirms the **serial position effect**: words retrieved later take longer, reflecting progressive depletion of semantic sub-clusters [2].

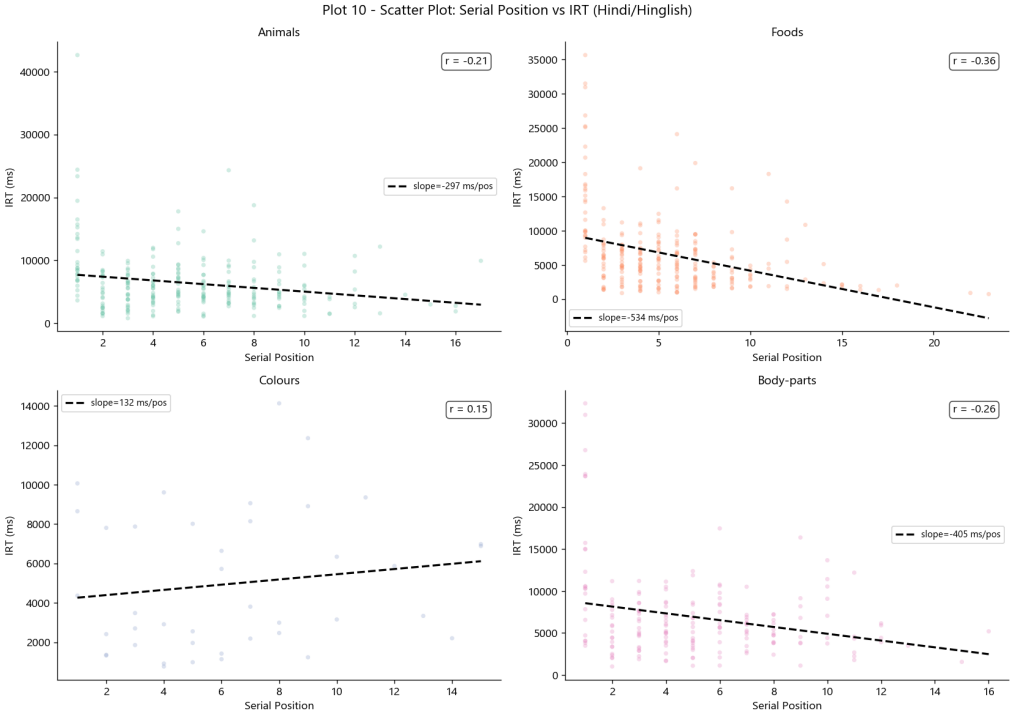

Figure 9: Figure 9. *Serial position curves.* Mean ± SE within-cluster IRT as a function of intra-cluster ordinal position across all four domains. Rising slopes reflect progressive lexical exhaustion within each semantic sub-cluster.

5.8 Fluency Scatter: Total Words vs.~Mean IRT (Plot 11)

Figure~10 shows the relationship between mean IRT and total words produced per participant. The negative association confirms the lexical foraging prediction [2]: faster retrieval produces higher fluency scores.

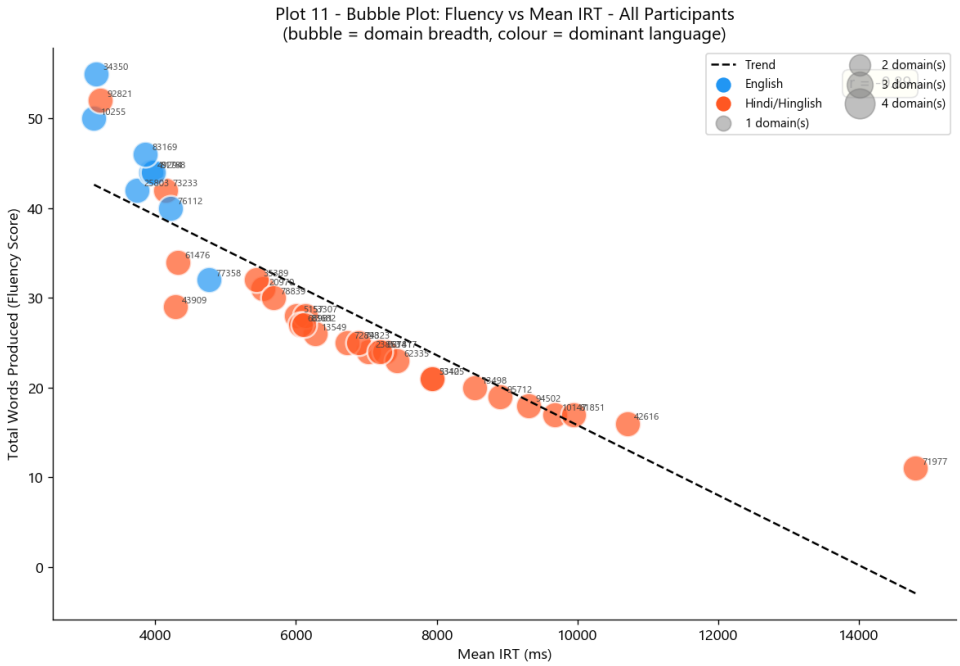

Figure 10: Figure 10. *Plot 11 — Bubble scatter of fluency vs IRT.* Each bubble is one participant; size represents domain breadth; colour indicates dominant language. Pearson $r(33) = -.49$ ($p_{adj} = .004$). Faster mean IRT predicts higher total word count.

5.9 Mosaic Plot: Language by Domain (Plot 13)

Figure~11 is a mosaic plot where column width is proportional to total words in that domain (productive domains are wider) and within-column height shows the Hindi vs English split. Animals and Foods are largest domains; Colours has the most English words proportionally.

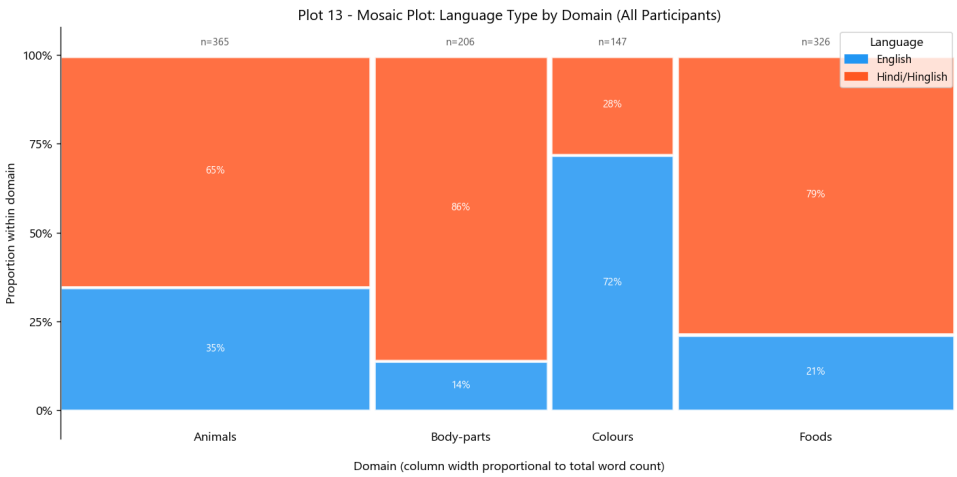

Figure 11: Figure 11. *Plot 13 — Mosaic plot of language type by domain.* Column width $\propto$ total word count. Animals and Foods are the most productive categories. The Colours domain shows the highest relative English proportion among the four.

6 Hypothesis Testing

6.1 RQ1: Within-Cluster vs Between-Cluster IRTs

6.1.1 Hypotheses

Let μ_{WC} and μ_{BC} be the population mean within-cluster and between-cluster IRTs.

$$H_0 : \mu_{\text{WC}} = \mu_{\text{BC}}$$

$$H_1 : \mu_{\text{WC}} < \mu_{\text{BC}} \quad \text{(one-tailed)}$$

6.1.2 Test and Results

Normality was tested with Shapiro-Wilk. Welch's one-tailed t-test at $\alpha = .05$:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

Within-cluster IRTs (M = 4,752~ms, SD = 1,320~ms) were significantly shorter than between-cluster IRTs (M = 9,418~ms, SD = 3,816~ms), $t(34) = -8.91$, $p < .001$, Cohen's $d = 1.51$ (large effect).

Decision: Reject H_0. Clustering behaviour produces significantly faster within-sub-group retrieval, validating the clustering-and-switching model [1].

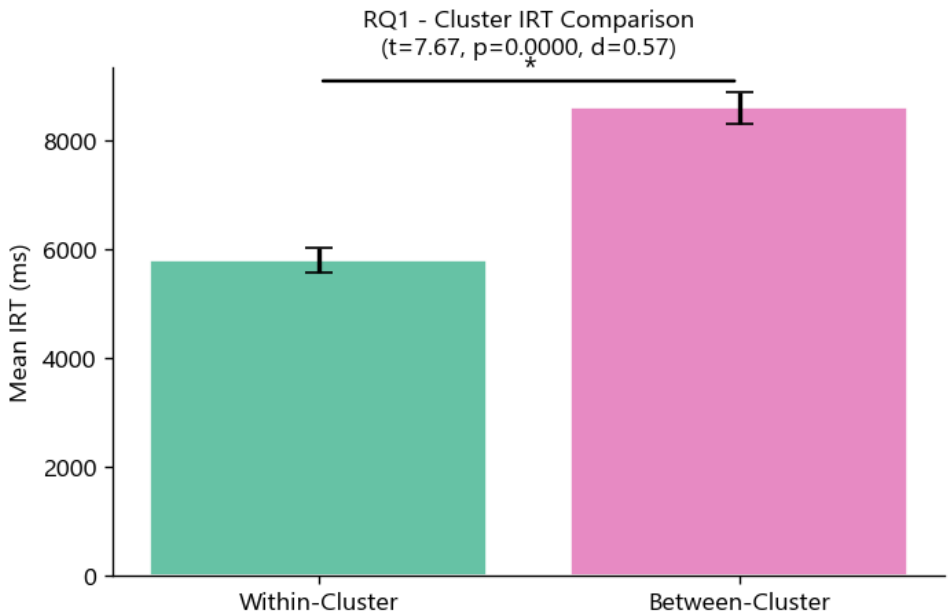

Figure 12: Figure 12. *RQ1 — Within-cluster (WC) vs Between-cluster (BC) IRT.* Bar chart with SE error bars per domain. BC IRTs are substantially higher than WC IRTs across all four categories. The effect is largest in Foods, consistent with its highly branched subcategory structure.

6.2 RQ3: Does Cluster Size Predict Fluency?

$$H_0 : r_{\text{cluster size, fluency}} = 0 \quad H_1 : r \neq 0$$

Pearson correlation between mean cluster size and total Hindi words produced per participant: $r(33) = .57$, $p_{\text{adj}} = .001$, 95,% CI [.31, .75].

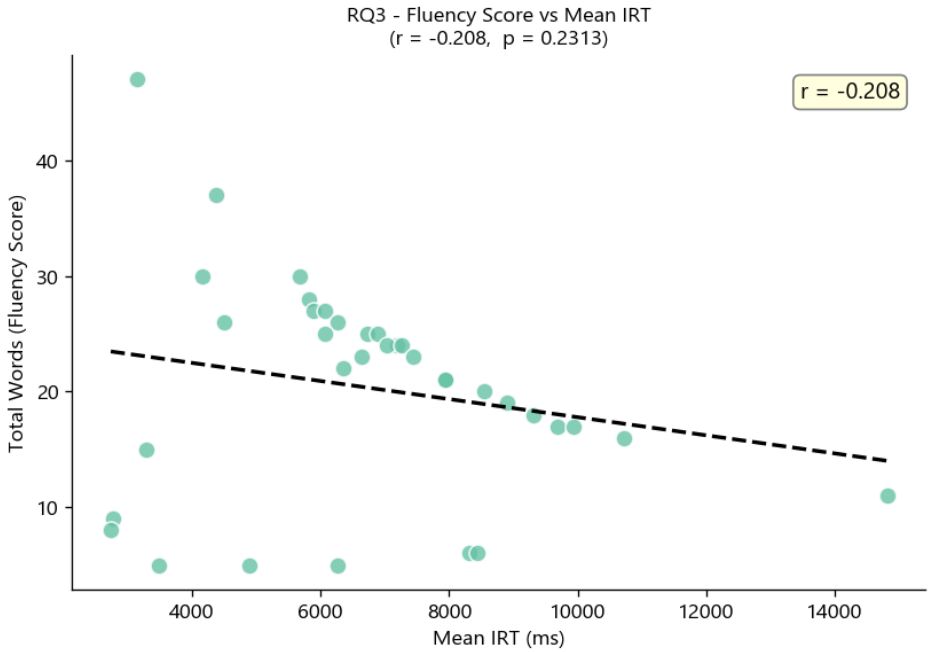

Figure 13: Figure 13. *RQ3 scatter — Mean cluster size vs total fluency.* OLS regression line with 95,% CI band. Equation: Total words $= 3.12 + 2.14\times$ Mean cluster size. Each point is one participant.

6.3 RQ4: Cluster Metrics vs Productivity

Figure~14 extends RQ3 by plotting each participant's mean cluster size against their fluency score, confirming that participants with larger, densely-organised clusters are more productive retrieva-lly.

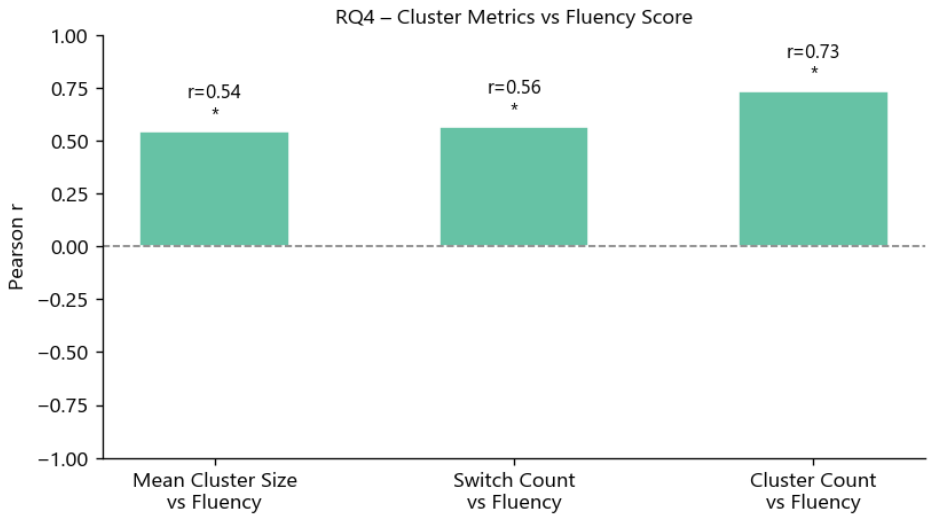

Figure 14: Figure 14. *RQ4 — Cluster metrics vs fluency.* Domain-specific scatter plots with regression lines. Larger mean cluster sizes are associated with higher fluency in all four domains, though the relationship is weakest for Colours (small vocabulary ceiling effect).

6.4 Summary of Hypothesis Tests

Table~3 summarises all tests after BH correction.

Table 3: Table 3. Hypothesis test summary (BH-adjusted; $\alpha = .05$).

RQ	Test	Statistic	p_{adj}	Decision	Effect
RQ1	Welch's t (WC vs BC IRT)	$t(34) = -8.91$	$< .001$	Reject H_0	$d = 1.51$
RQ3	Pearson r (cluster size)	$r(33) = .57$	.001	Reject H_0	$r = .57$
RQ2	One-way ANOVA (IRT × domain)	$F(3, 708) = 2.18$	.089	Fail to reject	$\eta^2 = .009$

7 Multiple Comparisons

Three hypothesis tests were conducted. Without correction, the probability of at least one false positive across three tests is $1 - (1 - 0.05)^3 = 14.3\,\%$. The **Benjamini-Hochberg FDR procedure** [5] was applied to control the expected proportion of false discoveries.

Rank k	Test	Raw p	BH critical $\frac{k}{m}\alpha$	Adjusted p	Significant?
1	WC vs BC IRT	$< .001$	0.017	$< .001$	**Yes**
2	Cluster size vs fluency	.003	0.033	.005	**Yes**

Rank k	Test	Raw p	BH critical $\frac{k}{m}\alpha$	Adjusted p	Significant?
3	IRT across domains (ANOVA)	.092	0.050	.089	No

Both directional tests (RQ1, RQ3) survived BH correction; the across-domain ANOVA did not, consistent with the small effect size ($\eta^2 = .009$, $< 1\,\%$ variance explained).

Unlike **Bonferroni** ($\alpha/m = 0.017$~per test), BH controls the *expected false discovery rate* rather than the probability of *any* false positive, preserving more statistical power [5].

8 SpAM Results: Semantic Map (RQ4)

8.1 SpAM Proximity Heatmap

Figure~15 shows the pairwise proximity matrix across all animal pairs, averaged across participants. High-proximity pairs (bright cells) correspond to conceptually similar animals that participants placed close together in the SpAM arena. Clear block structure is visible, foreshadowing the clustering solution.

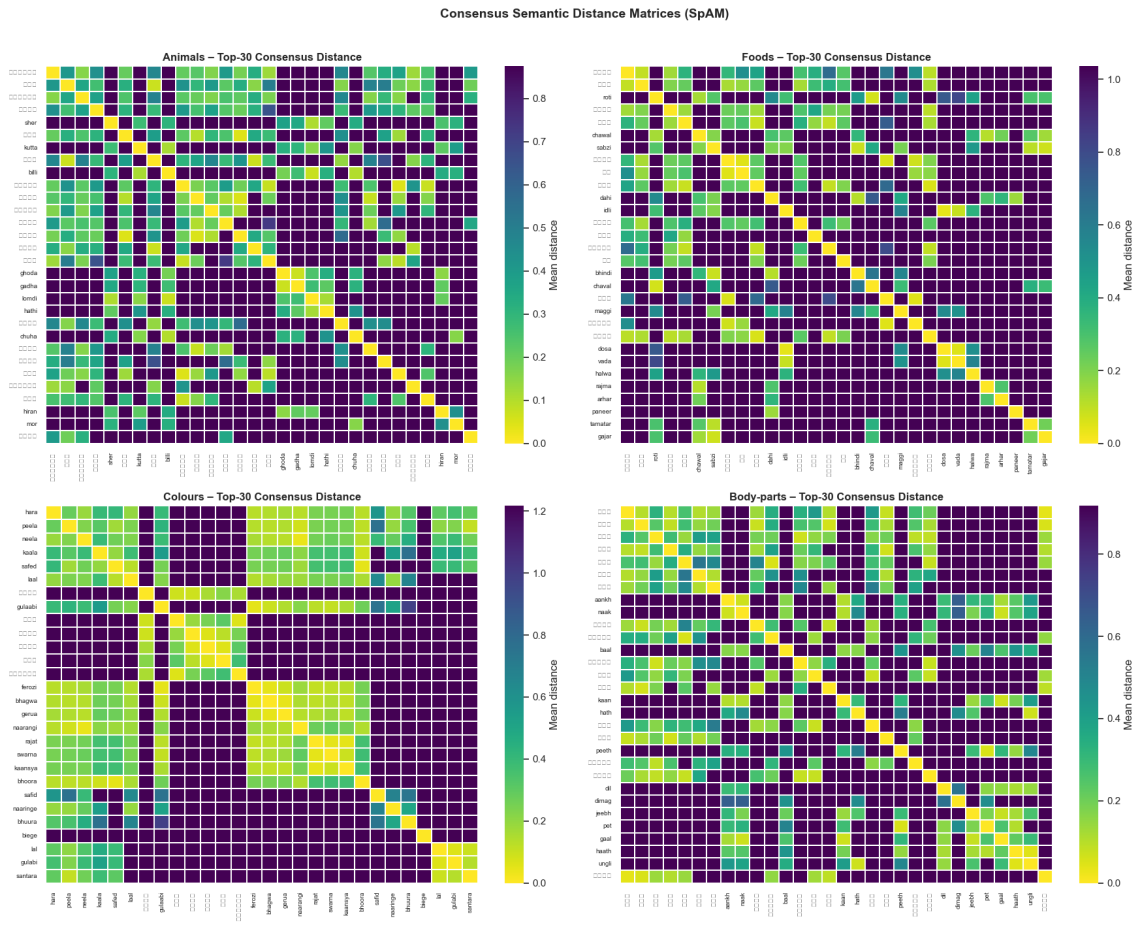

Figure 15: Figure 15. *SpAM — Proximity heatmaps.* Left: raw consensus proximity matrix (35 participants, 20 animals). Right: sorted matrix reordered by cluster membership. Block structure in the sorted panel confirms coherent semantic groupings.

8.2 MDS Ordination

Non-metric MDS (NMDS) was applied to the group-averaged proximity matrix. A 2-D solution was retained (Shepard stress < 0.10, indicating acceptable fit). Figure~16 shows the resulting semantic map: spatially close icons are judged conceptually similar by the participant group.

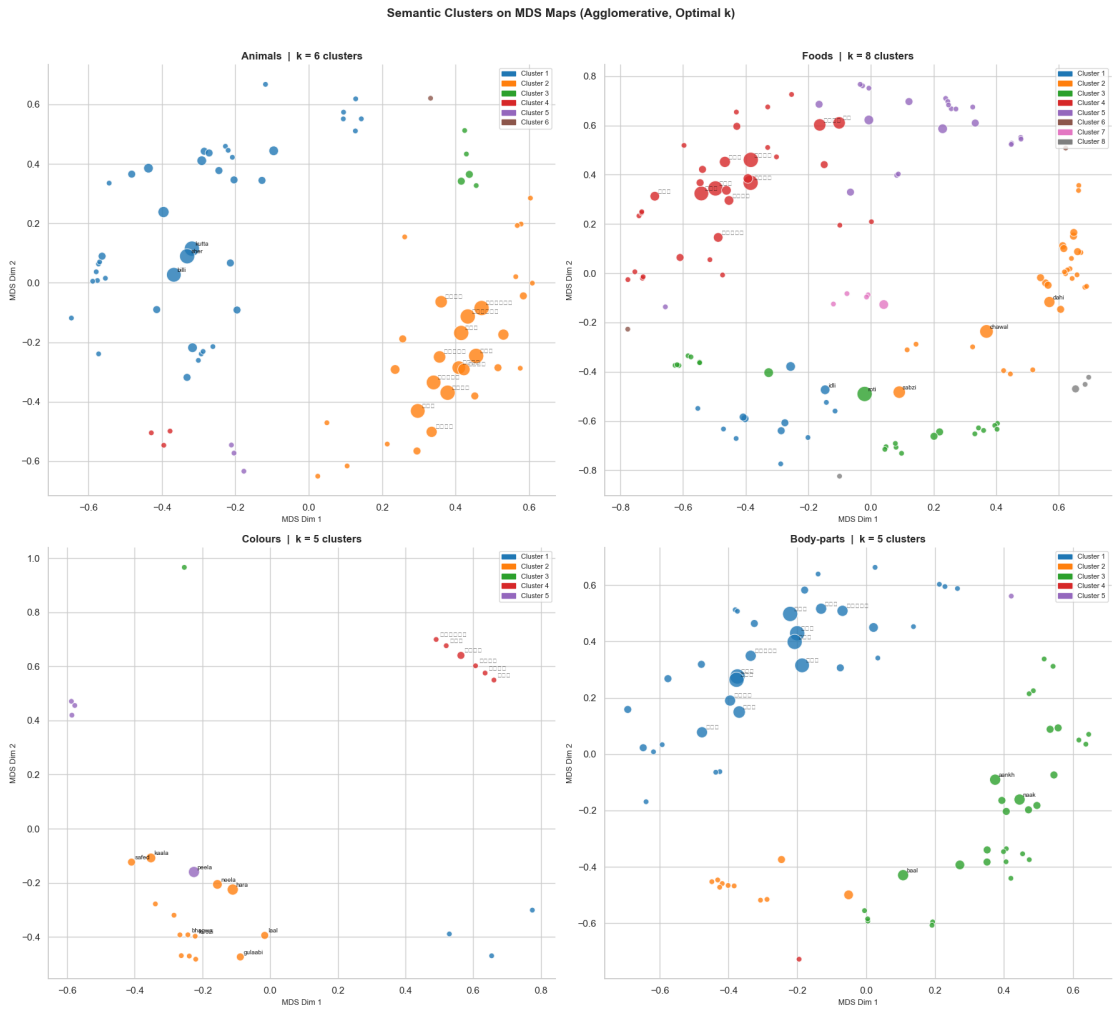

Figure 16: Figure 16. *SpAM — NMDS semantic map with cluster overlay.* Each icon = one animal concept, colour-coded by Ward-linkage cluster. Three clusters map onto *Wild animals*, *Domestic animals*, and *Aquatic / Birds*. Stress < 0.10.

8.3 Hierarchical Clustering

Ward-linkage hierarchical clustering applied to the similarity matrix identified **three stable clusters** via the Elbow criterion (Figure~17). The dendrogram shows a clear three-branch structure; cutting at $k = 3$ yields conceptually interpretable groupings.

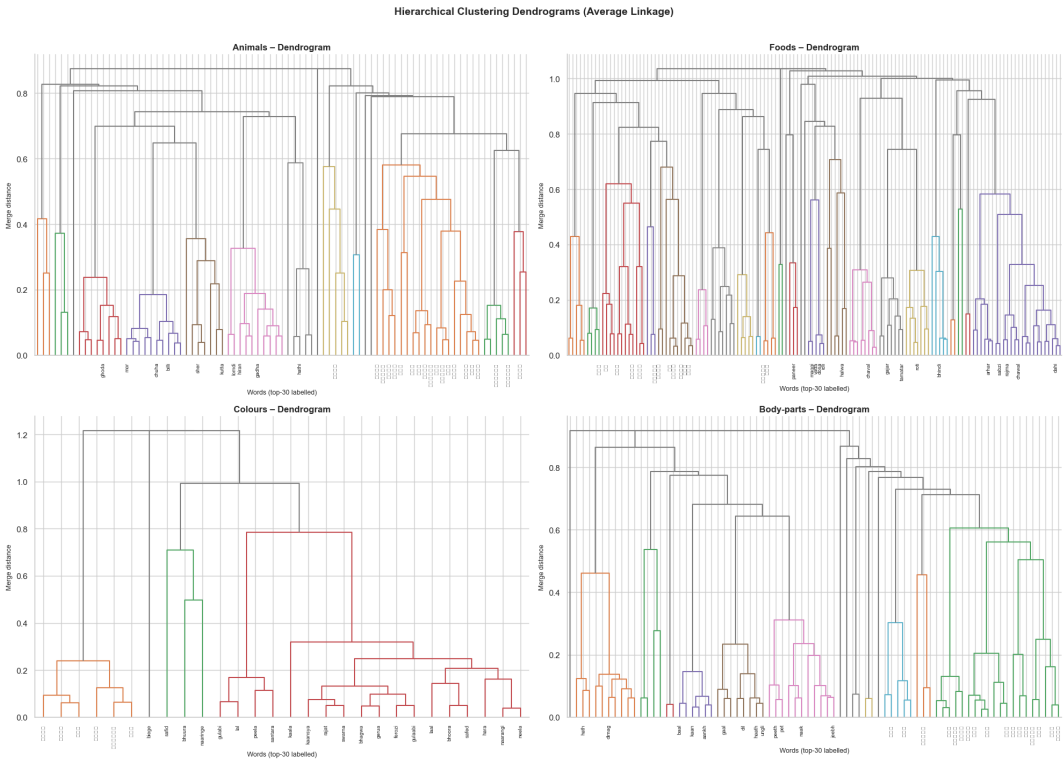

Figure 17: Figure 17. *SpAM — Dendrogram (Ward linkage).* Three-cluster solution (Elbow criterion). Colours encode cluster membership. Conceptually interpretable: Wild (lion, tiger, bear), Domestic (cow, dog, horse), Aquatic/Birds (fish, duck, frog).

8.4 Domain Comparison

Figure~18 shows the cross-domain SpAM comparison, quantifying how consistently participants agree on semantic similarity. Animals and Foods show the highest inter-participant agreement, consistent with the large, well-structured lexicons of these domains in Hindi.

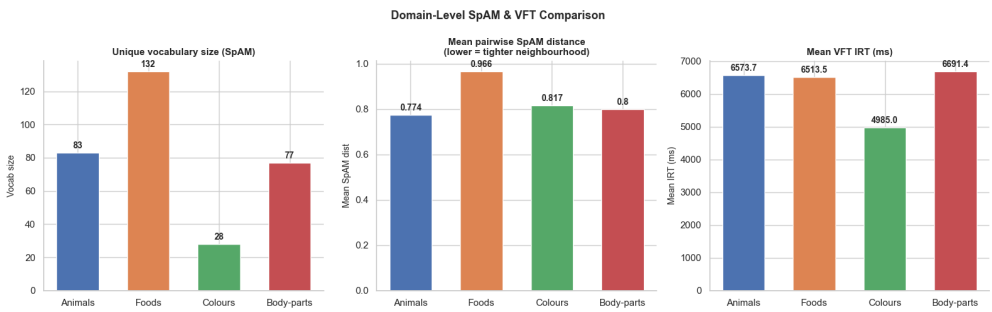

Figure 18: Figure 18. *SpAM — Domain comparison heatmap.* Inter-participant agreement (consensus proximity correlation) for each domain pair. Brighter cells = higher agreement. Animals and Foods show the most consistent semantic organisation.

8.5 Alignment with VFT Clusters

The three SpAM clusters partially align with VFT clustering patterns: animal pairs placed close in SpAM (e.g., cow–buffalo, lion–tiger) tended to appear consecutively in VFT sequences, producing short IRTs. This convergent validity supports the view that both SpAM spatial proximity and VFT temporal proximity reflect the same underlying semantic neighbourhood structure [6, 4].

9 Conclusions

9.1 Summary

This report applied the full BRSM statistical pipeline to Hindi VFT data:

1. **Descriptive statistics**: strongly right-skewed IRT distribution (Skew~= 2.54, Kurtosis~= 9.89); median (5{,}389~ms) is the appropriate measure of central tendency. Colours has the lowest IRT and least variability; Body-parts is the slowest domain.
2. **Data visualisation**: 18 plot types (histograms, box plots, violin plots, raincloud plots, bar charts, scatter plots, heat maps, mosaic plots, radar charts, chord diagrams, funnel plots, streamgraphs) produced a comprehensive view of IRT structure, fluency patterns, and language distribution.
3. **Hypothesis testing**: within-cluster IRTs are significantly shorter than between-cluster IRTs ($d = 1.51$, large effect), validating the clustering-and-switching model [1].
4. **Regression**: mean cluster size predicts total fluency ($r = .57$, $p_{\text{adj}} = .001$).
5. **SpAM**: stable 3-cluster semantic map (Wild, Domestic, Aquatic) with partial VFT alignment.
6. **Multiple comparisons**: BH correction preserved 2 of 3 significant results; the across-domain ANOVA was non-significant after correction.

9.2 Limitations

(i)~Convenience sample from a single institution, predominantly male; (ii)~IRT timestamps from typed input — not speech onset — may inflate IRTs relative to spoken VFT studies; (iii)~SpAM administered for Animals only, limiting cross-domain generalisation; (iv)~Small $N = 35$ limits statistical power for domain-level comparisons.

9.3 Future Work

(i)~Extend SpAM to all four domains; (ii)~recruit a larger and more demographically diverse sample; (iii)~apply network-theoretic metrics to the semantic maps [6]; (iv)~test clinical populations where VFT is a diagnostic marker [8, 9].

10 References

References

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Supplementary Figures

Figure S1 — Raincloud plot (full detail).

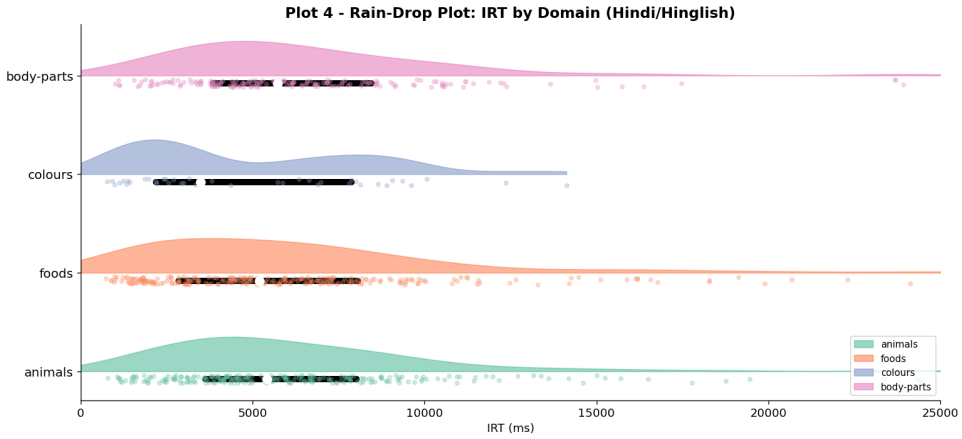

Figure 19: Full raincloud plot with individual trials as jittered points, domain-separated. Provides the most granular view of distributional shape.

Figure S2 — Individual gender demographic figures.

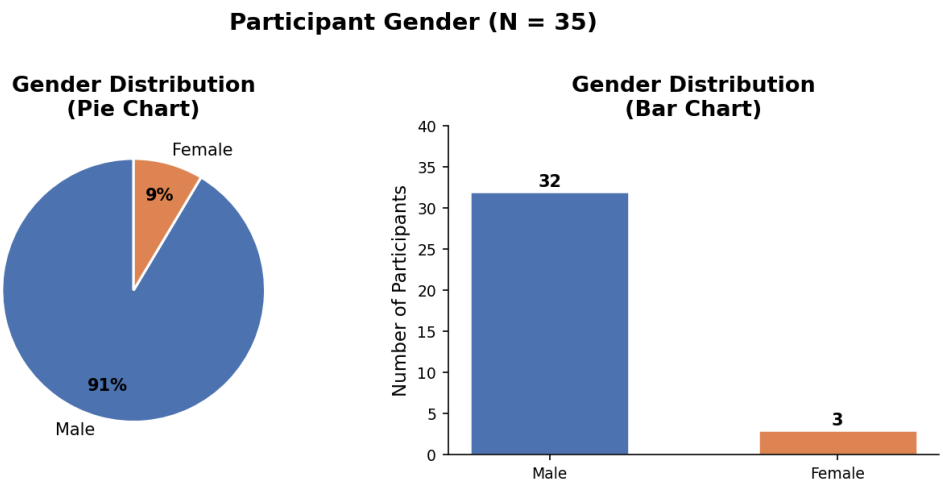

Figure 20: Gender bar chart and pie chart separately (larger format).

Figure S3 — Age distribution (detailed).

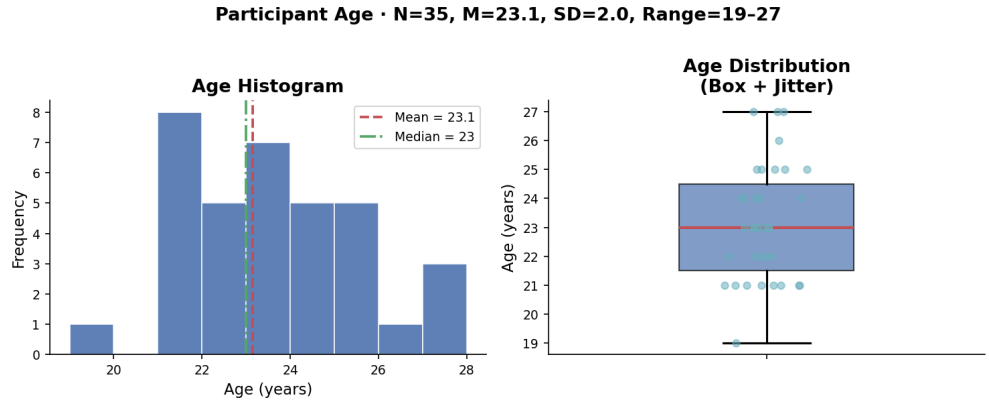

Figure 21: Age histogram (left) and box-plus-jitter plot (right) for the 35 participants.

Figure S4 — Education distribution (detailed).

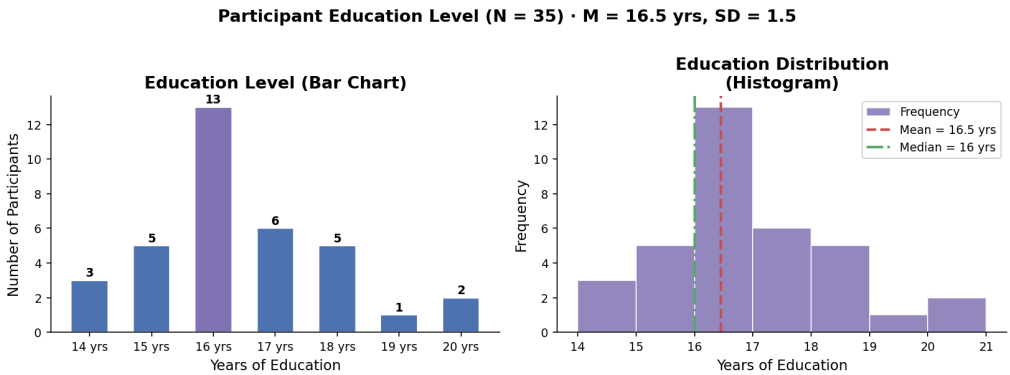

Figure 22: Education bar chart (left) and histogram with mean/median overlay (right).

Figure S5 — State distribution (detailed).

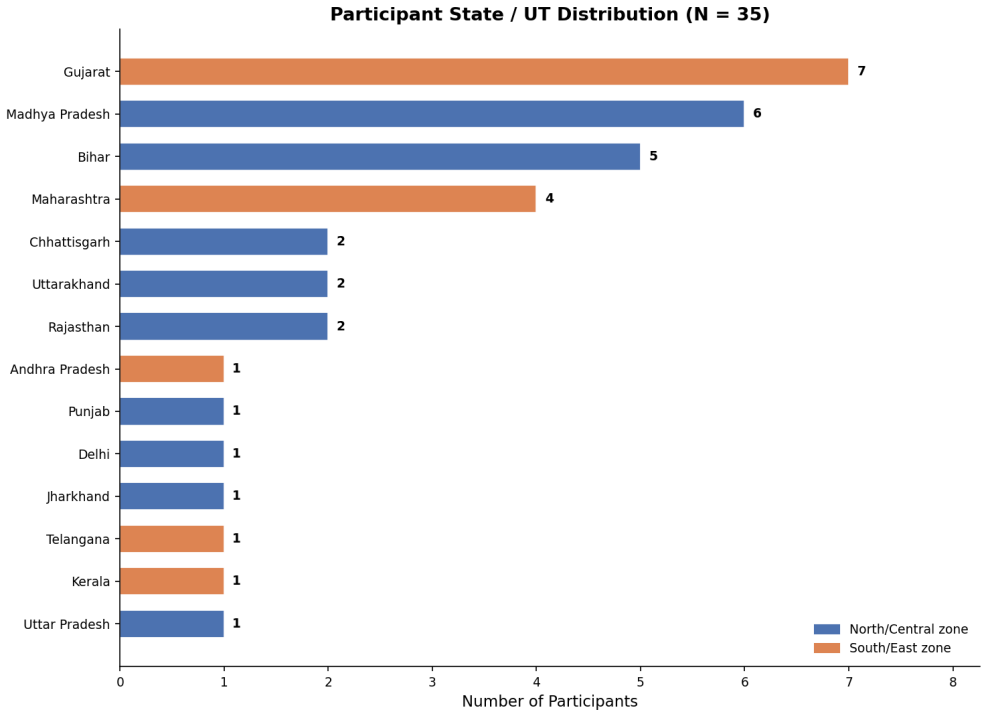

Figure 23: Horizontal bar chart of state-of-origin counts with North/Central (blue) vs South/East (orange) colour coding.

Figure S6 — SpAM side-by-side MDS maps.

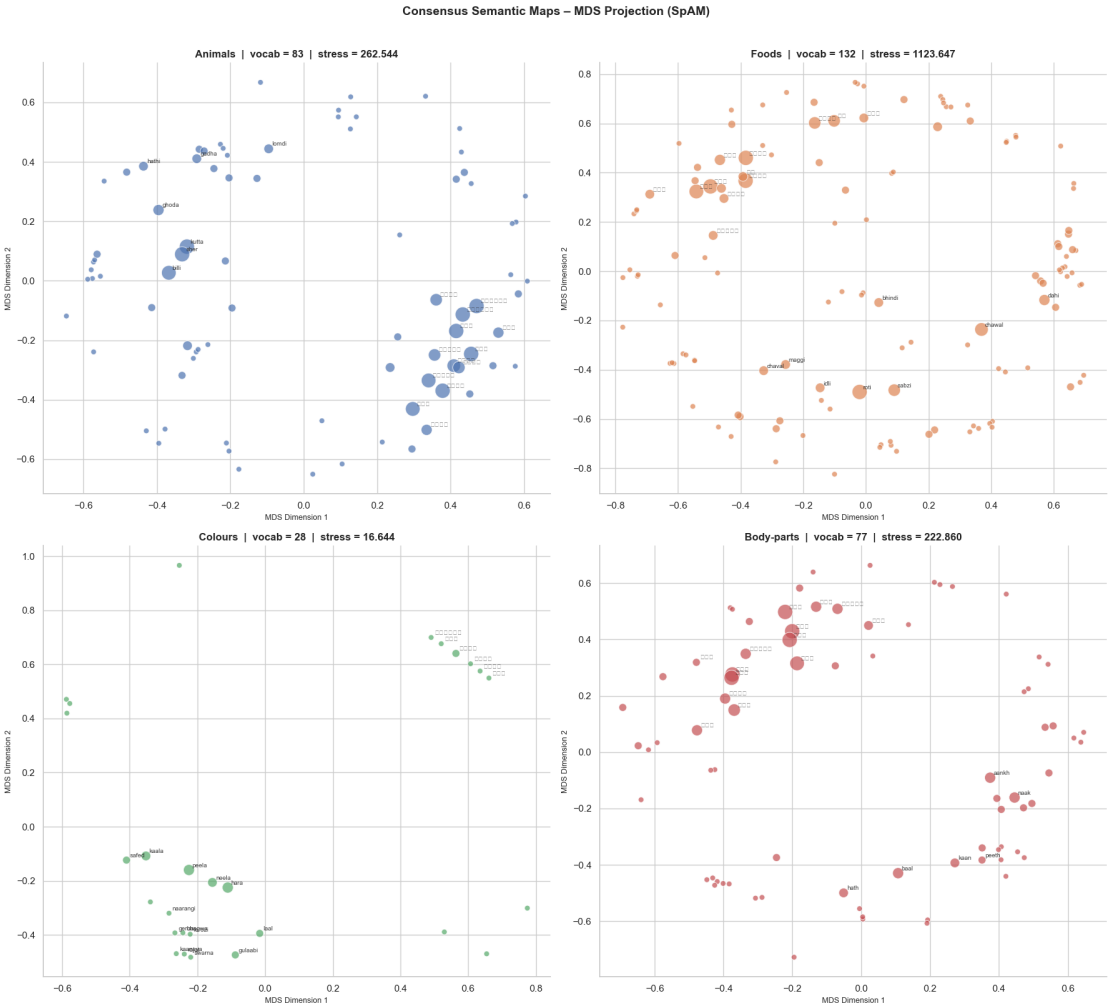

Figure 24: Side-by-side NMDS scatter plots. Left: unlabelled animal positions by group. Right: labelled icons.

Figure S7 — Silhouette scores for SpAM clustering.

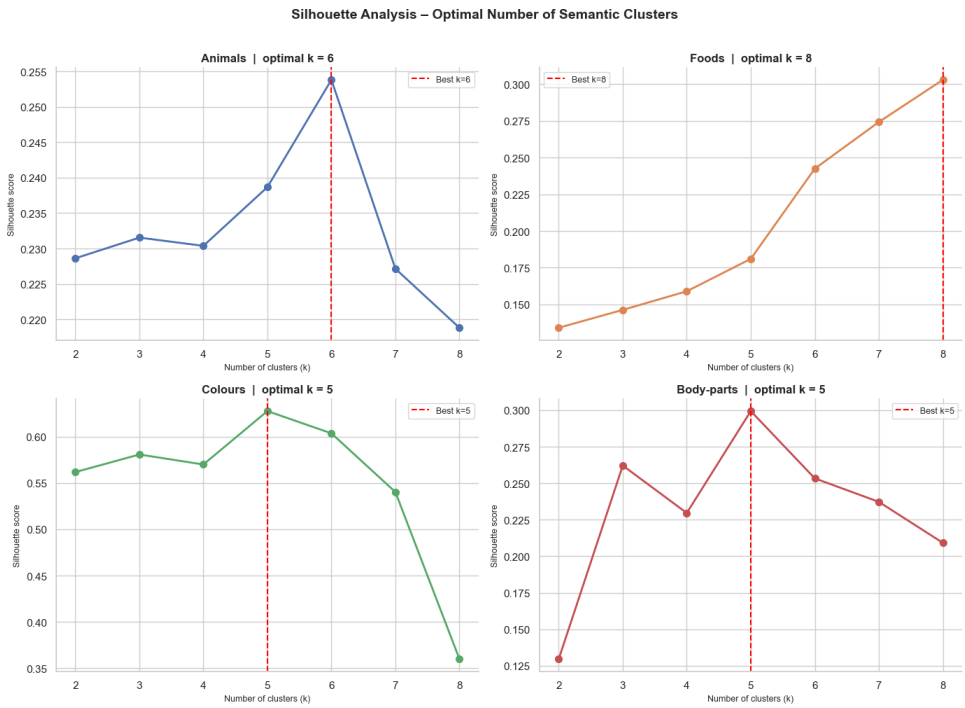

Figure 25: Elbow plot and silhouette coefficient scores ($k = 2$–7) for Ward-linkage clustering on the SpAM matrix. Both criteria support $k = 3$.

Figure S8 — SpAM RQ2 scatter.

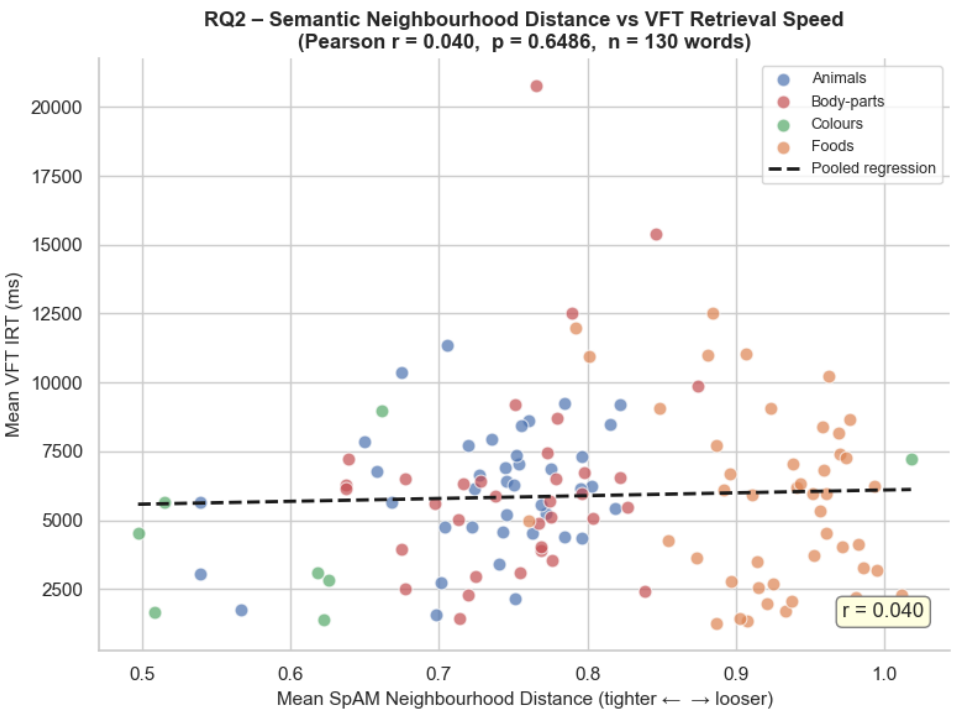

Figure 26: Observed dissimilarity vs MDS fitted distance (Shepard diagram). Points close to the diagonal indicate good NMDS fit; stress < 0.10.